

```

#include <stdio.h>

int mutex = 1;
int wrt = 1;
int readcount = 0;

int wait(int s)
{
    return --s;
}

int signal(int s)
{
    return ++s;
}

int main()
{
    int choice;

    while(1)
    {
        printf("\n1. Reader\n2. Writer\n3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                mutex = wait(mutex);
                readcount++;

                if(readcount == 1)
                    wrt = wait(wrt);

                mutex = signal(mutex);

                printf("Reader is reading...\n");

                mutex = wait(mutex);
                readcount--;
            }
        }
    }

```

```

        if(readcount == 0)
            wrt = signal(wrt);

        mutex = signal(mutex);
        break;

    case 2:
        if(wrt == 1)
        {
            wrt = wait(wrt);

            printf("Writer is writing...\n");

            wrt = signal(wrt);
        }
        else
        {
            printf("Writer is waiting...\n");
        }
        break;

    case 3:
        return 0;

    default:
        printf("Invalid Choice\n");
    }
}

```

OUTPUT:

1. Reader
2. Writer
3. Exit

Enter your choice: 1

Reader is reading...

1. Reader
2. Writer
3. Exit

Enter your choice: 1

Reader is reading...

1. Reader

2. Writer

3. Exit

Enter your choice: 2

Writer is writing...

1. Reader

2. Writer

3. Exit

Enter your choice: 3